

Eventless Payload-Adaptive NMPC for Quadrotors

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Abstract—A release command does not guarantee that a payload has physically detached, so a controller that clears its payload model on that command can fly a loaded vehicle as if it were empty. We present an eventless payload-model interface for quadrotor nonlinear model predictive control (NMPC) that never treats attach or release commands as evidence. A moving horizon estimator (MHE) [1] estimates composite mass and the horizontal first mass moment $s = m_P r_{xy}$ from odometry and a rotor-speed thrust proxy that is not computed from the payload estimate; unlike a center-of-mass offset, s remains well defined as the payload vanishes. The inertia increment follows from the mass estimate and a weak vertical-arm prior, and health-gated, rate-limited updates drive both the NMPC model and body-rate scheduling. In PX4 [2] SITL with physically attached eccentric payloads, release confirmation completed in all 31 flights that obtained payload evidence and survived to the release command (31 of 40 attempts). Confirmation took a median 3.46 s when evidence arrived during the maneuver and 16.48 s when the vehicle first had to brake to hover. When physical detachment was delayed by 4 s after the command, a command-triggered baseline cleared its model early in 5/5 paired flights and the proposed interface in 0/5. These results are limited to simulation and the tested conditions.

I. INTRODUCTION

Autonomous aerial delivery and in-flight aerial manipulation subject a multirotor to abrupt changes in its own dynamics. Grasping or releasing a parcel changes three coupled quantities at once: the total mass m (which rescales the thrust-to-acceleration map and the hover throttle), the center of mass (a thrust that no longer passes through the composite CoM produces a constant trim torque $\tau \approx \Delta r \times f$, significant even for small offsets because $f \approx mg$ is large), and the rotational inertia (which slows the attitude response and effectively lowers the inner-loop bandwidth). The first channel is an outer-loop (position/thrust) problem; the latter two are inner-loop (attitude/rate) problems. A controller that models the payload only as a scalar mass will mistrack or, for sufficiently heavy eccentric loads, destabilize.

Explicit parameter estimation (EKF/MHE of mass, CoM, inertia) is interpretable and can feed model-based constraints, but converges slowly across an abrupt change; *lumped-disturbance adaptation* (L1-adaptive NMPC, INDI) reacts fast but yields no physical model; *learning-based* estimators typically need rich sensing or ground-truth data. We take the explicit route deliberately and accept its latency: the proposed interface trades release-confirmation delay for semantics that never run ahead of physical evidence and for named physical parameters.

The central question of this work is whether the complete grasp-carry-release lifecycle can adapt its payload model without trusting a commanded or externally supplied payload

event. Our answer rests on two observations. First, payload mass and first moment can be estimated continuously, with the inertia increment following algebraically from the mass estimate and a weak geometric prior, without consuming a gripper command, attachment state, or external event notification. Second, rotor-speed telemetry provides a collective-thrust and body-torque proxy that is *not computed from* the online payload estimate, under a calibrated rotor thrust model; residuals formed from it therefore do not inherit a mass error through the controller’s own command. Combining the two, the estimator needs no external event signal and no measurement of the horizontal attachment geometry.

Contributions.

- 1) **A first-moment payload interface.** Estimating mass together with the horizontal first mass moment—the standard linear parameterization of rigid-body inertial parameters [3]—keeps the payload geometry well defined through release, where a CoM offset is not, and lets one health-gated $m/s/J$ frame drive bounded NMPC model updates and continuous body-rate scheduling (§III–IV).
- 2) **Release semantics that do not run ahead of evidence:** the controller clears its payload model only after the estimator’s no-payload confidence persists, never on its own release command, and enters UNRESOLVED and brakes to hover when evidence is missing (§IV).

We evaluate the complete interface in PX4 SITL, reporting every attempt, the confirmation path each flight took, and a paired delayed-detachment fault injection (§VI–A). The runtime estimator consumes only motor speeds and odometry; the sole geometric quantity not identified online is a weak vertical-arm prior r_z , which is structurally unobservable from the thrust-moment channel.

II. RELATED WORK

Agile quadrotor flight. Geometric tracking on SE(3) [4] and time-optimal flight [5] assume known, fixed inertial parameters, which a payload event invalidates mid-flight.

Aerial manipulation and transport. Aerial robots that grasp, carry and release objects are surveyed in [6]; dynamic aerial grasping, with explicitly modeled gripper dynamics, was demonstrated in [7]. Load transportation, by grasping or cable suspension, is surveyed in [8]. Ours is the transport-and-release subproblem: the payload is rigidly attached, so the controller must handle the resulting change in mass, inertia and centre of mass.

Control under changing payload. Adaptive and robust designs—L1-adaptive NMPC [9], INDI [10]—treat the load

as a lumped disturbance and are the standard baselines for payload-carrying flight; L1-NMPC in particular reports large tracking-error reductions under unknown load on aggressive trajectories. A parallel line learns the residual dynamics instead: [11] fits unmodeled aerodynamic effects with Gaussian processes inside the MPC, which improves tracking without yielding a named physical parameter either. Slung-load control addresses the swinging-payload case explicitly [12], exploiting differential flatness of the cable-suspended system; a rigidly attached payload instead changes the vehicle's own inertial parameters. A related proprioceptive line estimates the external wrench directly with momentum-based observers, enabling collision detection and interaction control [13]. All of these recover a *force/torque*, not a physical parameter: they cannot, e.g., update a thrust-margin constraint or an inner-loop gain schedule from the estimate.

MHE and online parameter estimation. Filter- and optimization-based estimators recover mass, inertia, and geometric parameters [14]–[16], and the mass and CoM of a grasped payload have been estimated for aerial manipulation [17]. Writing the CoM as a first mass moment makes inertial parameters enter the dynamics linearly, a standard device in manipulator load identification [3]; we use it because it remains well defined when the payload vanishes. Standard MHE performance is highly sensitive to the process/measurement weight matrices; neural MHE (NeuroMHE) [18] learns to output these weights online, resolving the bias–variance trade-off but estimating a lumped disturbance force rather than explicit named parameters.

Residual-based change detection. Residual-based fault/change detection (CUSUM and related methods [19]) motivates the evidence checks used by the payload lifecycle state machine. Our positioning: we do not learn dynamics and add no sensing; every deployed signal derives from motor speeds and odometry. This is a deliberate departure from learning-based MPC [20] and the broader safe-learning agenda [21], where a learned component absorbs the model mismatch; here the mismatch is resolved into named parameters that downstream constraints can consume.

III. SYSTEM OVERVIEW AND PROBLEM SETUP

Architecture (Fig. 1). Three layers: Gazebo physics (with a magnetic DetachableJoint gripper producing genuine inertia/CoM changes), the PX4 inner-loop rate controller, and a ROS 2 outer loop (NMPC + MHE). The outer loop publishes a collective thrust and a body-rate setpoint; PX4 tracks the body rate.

Dynamics and parameterization. The NMPC model carries the runtime parameter vector $p = [x_{\text{ref}}, m, \Delta J, c_{xy}]$ (extended with a lumped disturbance only for the L1 baseline). Thrust acting at the geometric center about the composite CoM produces $\tau_{\text{com}} = [-c_y T, +c_x T, 0]$. The NMPC solves

$$\begin{aligned} \min_{u_{0:N-1}} \quad & \sum_{k=0}^{N-1} (\|e_k\|_Q^2 + \|u_k - \bar{u}(p)\|_R^2) + \|e_N\|_Q^2 \\ \text{s.t.} \quad & x_{k+1} = F(x_k, u_k, p), \quad x_0 = \hat{x}, \quad u_k \in \mathcal{U}, \end{aligned} \quad (1)$$

where $x \in \mathbb{R}^{13}$ stacks position, velocity, attitude quaternion and body rate, $u = [T, \tau^\top]^\top$, e_k stacks the position, velocity, quaternion-vector and rate errors to the reference, and F is one RK4 step ($\Delta t = 0.05$ s, $N = 20$). The input reference $\bar{u}(p) = [mg, c_y mg, -c_x mg, 0]^\top$ is the live hover input including the eccentric trim torque. Q and R follow Bryson's rule with tolerances 0.1 m, 0.3 m/s, 0.05, 0.3 rad/s, 2 N and the torque bounds; Q_N scales the position, attitude and rate blocks of Q by 20, 2 and 0.2. $\mathcal{U}$ is the actuator envelope $T \in [1.27, 31.35]$ N, $|\tau_x|, |\tau_y| \leq 0.5$ N m, $|\tau_z| \leq 0.2$ N m. The deployed MHE augments both composite mass and the horizontal first mass moment,

$$\begin{aligned} x_{\text{MHE}} &= [x^\top \quad m \quad s_x \quad s_y]^\top \in \mathbb{R}^{16}, \\ s &= m p r_{xy}, \end{aligned} \quad (2)$$

with $\dot{m} = 0$ and $\dot{s} = 0$ within each window ($N = 20$, $\Delta t = 0.1$ s). It solves

$$\begin{aligned} \min_{x, w} \quad & \|x_0 - \bar{x}_0\|_{W_0}^2 \\ & + \sum_{j=0}^{N-1} \beta_j (\|y_j - C x_j\|_{W_y}^2 + \|w_j\|_{W_w}^2) \\ \text{s.t.} \quad & x_{j+1} = F_M(x_j, u_j^{\text{phys}}) + G w_j, \\ & m \in [0.95 m_B, 5 \text{ kg}], \quad |s_x|, |s_y| \leq 1.5 \text{ kg m}, \end{aligned} \quad (3)$$

over $x_{0:N}$ and $w_{0:N-1}$, where y_j is the odometry and C selects the 13 vehicle states, $u^{\text{phys}} = [T_{\text{phys}}, \tau_{\text{phys}}^\top]^\top$ is reconstructed from rotor speeds (below), and $G = [I_{13}, 0]^\top$ keeps m and s noise-free. W_y is the inverse squared odometry noise (0.02 m, 0.05 m/s, 0.01, 0.02 rad/s); $W_w = \text{diag}(10^4 I_6, 10^5 I_4, 10^3 I_3)$ deliberately trusts the dynamics. The arrival cost $W_0 = \text{diag}(10^3 I_6, 10^4 I_4, 10^2 I_3, 0.1, 100 I_2)$ is centred on the previous window's shifted solution and is nearly uninformative in m ($\sigma \approx 3.2$ kg) and s ($\sigma = 0.1$ kg m). In the reported campaign $\beta_j = 1$, except that stages preceding a self-detected T_{phys} step (4.4 N from a slow baseline for two frames) are scaled by 10^{-4} until they leave the window; no external event enters this schedule. The published geometry is reconstructed as

$$\begin{aligned} c_{xy} &= \frac{s}{m}, \\ \mu &= \frac{m_B \max\{m - m_B, 0\}}{m}, \\ \Delta J_{xx} &= \Delta J_{yy} = \mu r_z^2, \end{aligned} \quad (4)$$

where r_z is a weak prior from the payload rack. The small $O(r_{xy}^2)$ diagonal terms and yaw increment are omitted in the deployed interface.

Estimate-independent thrust proxy. We reconstruct the collective thrust $T_{\text{phys}} = k \sum_i \omega_i^2$ and the body torques from rotor speeds and arm geometry. The quadratic coefficient k is a static-thrust constant; it is known to lose validity under large induced inflow during fast translation [22], which we do not model. Feeding the NMPC's *commanded* thrust back into the estimator would instead close an estimator–controller loop: that command is computed from $\hat{m}$ itself (its cost anchors T near $\hat{m}g$), and any inversion error, saturation, or motor lag between command and rotor is invisible to the estimator, so a

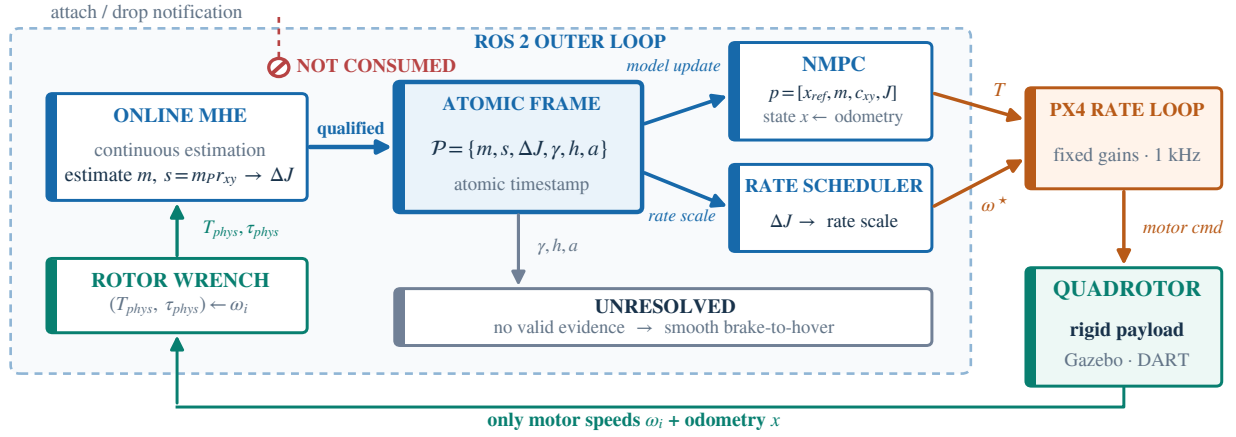


Fig. 1. System architecture and information flow. The full pipeline consumes only motor speeds and odometry; one atomic, health-gated $m/s/J$ frame drives both the NMPC model and continuous rate-command scheduling. No attach/drop notification enters this path.

mass error can persist self-consistently rather than appear as a residual. The rotor-speed proxy is taken downstream of those stages and is not computed from $\hat{m}$; it does depend on the calibrated k , and in our simulator the plant generates thrust with the same square law (§VII).

Thrust-to-throttle inversion. The offboard collective-thrust command is inverted to a normalized rotor command through the motor model $T = k\omega^2$. Provided that the forward thrust reconstruction and inverse command mapping use the same k , steady-state closed-loop tracking is generally not directly affected by its absolute value.

Continuous inner-loop scheduling. Model consistency in the NMPC cannot by itself restore the PX4 rate loop’s lost bandwidth. The deployed path therefore maps the live inertia increment to a body-rate-command scale, low-pass filters it, applies target hysteresis and asymmetric rise/recovery slew limits, and preserves 5% command headroom before the final saturation. It does *not* rewrite PX4 parameters. An attach-time inertia-only bootstrap uses the rack’s 0.30 kg maximum payload for immediate safety, then fades out after persistent coherent MHE evidence; neither payload mass nor s is injected by this bootstrap, and the drop path remains estimate-only.

IV. ONLINE ESTIMATION AND EVENTLESS ADAPTATION

A. Continuous payload interface and autonomous lifecycle

The MHE publishes one atomic frame

$$\mathcal{P} = \{m, s_x, s_y, c_x, c_y, \Delta J_{xx}, \Delta J_{yy}, \Delta J_{zz}, J_{xx}, J_{yy}, J_{zz}, \gamma, h, a\}, \quad (5)$$

where γ is no-payload confidence, h is a health bit and a is solution age. Health requires a successful latest solve and fresh odometry, motor-speed, and known-input streams. The NMPC additionally checks local receive age; an unhealthy or stale frame freezes the last accepted model and breaks the

confidence persistence timer. Valid frames pass through a low-pass filter and separate slew limits for m , s , and ΔJ before every NMPC solve.

Payload presence is an estimator-owned state, distinct from any externally signalled attach state. A hysteretic $\text{EMPTY} \leftrightarrow \text{LOADED}$ machine uses sustained payload-mass evidence, with a directional residual as a fast path, and a separate latch records that the present flight reliably entered LOADED . This latch prevents an empty-flight confidence value from consuming the later drop confirmation and prevents a transient mass dip during a figure-eight from disarming the true release path.

No-payload confidence is the product of smooth empty scores for payload quantity, first moment, and inertia. A nonzero moment therefore vetoes a false empty decision even if the mass estimate touches its lower bound. On accepted release evidence, the estimator switches only the *published target* of s to zero; the optimization state is not overwritten. The output filter consequently produces a continuous decay rather than a geometric step. The controller marks a planned drop complete, resets its L1 state, and finishes recovery only after healthy confidence remains above 0.90 for 1.0 s; issuing the gripper command itself does none of these things.

B. Why the first mass moment is the essential state

Estimating CoM offset alone is ill-conditioned at release: physically $s = m_P r_{xy} \rightarrow 0$, whereas r_{xy} itself is undefined as $m_P \rightarrow 0$. In an earlier mass-only formulation, the optimizer could suppress a phantom eccentricity only by driving total mass below the bare-airframe value; 14 of 55 post-release samples then pinned at the lower bound. Augmenting s supplies the rotational channel directly and permits $c_{xy} = s/m$ without knowing the grasp-dependent horizontal offset. A frozen-window inertia coefficient prevents the optimizer from using instantaneous mass as a free inertia knob, while the weak r_z prior retains the dominant parallel-axis term. It is the smallest structure we found that represents payload existence, trim moment, and roll/pitch inertia without an external attachment geometry measurement.

Estimator declaration and controller confirmation. The controller clears its payload model through one route only: confirmation, i.e. $\gamma > 0.90$ held for 1.0 s on healthy, fresh frames. The estimator may additionally *declare* release, which switches the published target of s to zero and so speeds the rise of γ ; a declaration is neither necessary nor sufficient for confirmation. The final code has two declaration routes. (a) A *moment-gated detector* requires collapse of $\rho = \|s\|/\|s\|_{\text{ref}}$ below 0.30, where the loaded reference is formed from a healthy, low-dynamic, in-envelope interval and frozen before aggressive flight. It fires on quantity-empty plus moment-collapse evidence for five frames (slow path), or on moment collapse accompanied by a persistent signed thrust-residual step $\Delta T = \bar{T}_{\text{post}} - \bar{T}_{\text{pre}}$ over 0.3 s half-windows (moment-residual path). (b) A *mass-domain route* declares release after $m_P < 0.030$ kg for 20 frames, accumulated only in low-dynamic flight. Confirmation can also occur without any declaration; because the confidence’s moment score uses an absolute bound on $\|s\|$, a nonzero moment still vetoes it. Neither (b) nor confidence alone uses the frozen-reference ratio, so confirmations through them are not evidence for (a); Sec. VI-A reports how often each occurred. Moment evidence must come from a fresh, successful solve: re-anchoring after five consecutive solver failures initializes s to zero, and that value is not counted as collapse. During development we also tested a quantity-residual branch (a 0.9 N residual plus quantity-empty evidence, without moment evidence); it was removed after premature releases (Sec. VI-A). A recorded figure-eight transient reached -1.69 N, larger in magnitude than the -1.47 N step of the smallest payload, so residual amplitude alone cannot separate the cases. Online and offline evaluation call the same pure decision function, and an UNRESOLVED timeout is treated as unknown, never as proof of an empty vehicle.

Table I lists the lifecycle thresholds and where each value comes from. All were fixed in code before the final campaign began and were not retuned on it. Only ρ and the residual floor were selected against data from earlier flights; the remaining values are engineering choices whose closed-loop sensitivity we have not swept.

V. SIMULATION SETUP

PX4 SITL with Gazebo [23] (Harmonic), an x500 quadrotor, and a magnetic DetachableJoint gripper that produces genuine inertia and CoM changes.

The physical DetachableJoint gripper attaches a separate rigid body and therefore couples mass, inertia and CoM together. It supports the lifecycle, baseline, and safety claims in the paper; no wrench-emulated mass-step result is retained.

Task sequence (Fig. 2): approach (position control) $\rightarrow$ NMPC hand-off and warmup hover $\rightarrow$ controlled descent $\rightarrow$ active grasp enable $\rightarrow$ lift $\rightarrow$ figure-eight $\rightarrow$ phase-triggered release. We state the NMPC scope explicitly: it governs the grasp hover and the post-event adaptation phase; the approach controller is not part of the claim. Batches run headless, with

TABLE I
LIFECYCLE THRESHOLDS AND THEIR PROVENANCE. “EARLIER FLIGHTS” PREDATE, AND ARE DISJOINT FROM, THE REPORTED CAMPAIGN.

threshold	value	provenance
moment-collapse ratio ρ	0.30	leave-one-out cross-validation on 59 earlier flights; every fold chose 0.30 (0.20 succeeded in 48/59)
residual floor / persistence	0.9 N; 2 or 4 frames	below the 1.47 N weight of the smallest payload; on a recorded flight, figure-eight excursions above 0.9 N lasted ≤ 3 frames, the release step 4 frames
slow-path persistence	5 frames (0.5 s)	engineering choice
mass-domain release	$m_P < 0.030$ kg, 20 frames	between the loaded minimum (0.079 kg) and post-release maximum (0.007 kg) of earlier 0.15 kg flights
no-payload confidence	0.90 for 1.0 s	engineering choice
UNRESOLVED timeout / brake	12 s / 3 s	engineering choice

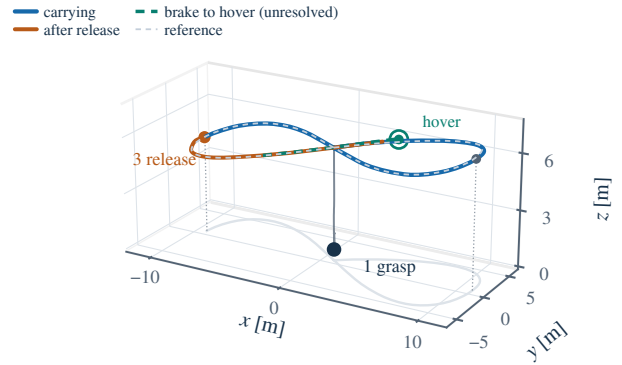


Fig. 2. Measured mission trajectory (simulator ground truth), two 0.15 kg/2 m/s flights of the same batch: grasp (1), lift, then carry along the figure-eight with ± 0.8 m vertical undulation (blue) against the nominal reference (pale dashed). After the release command (3) one flight clears the moment vote in maneuver and keeps tracking (orange); the other declares UNRESOLVED at its 12 s timeout and brakes to a hover (teal), confirming at 16.40 s. The two paths agree to within 0.12 m (median) before the timeout; loaded tracking RMS is 0.053 m, and 0.060 m after release.

a stand-in process supplying the ground-station heartbeat that PX4 requires.

Real-time budget. Both optimizers are acados [24] SQP solvers with merit backtracking and the HPIPM [25] QP solver. On one desktop core shared with Gazebo/PX4 SITL, the deployed 13-state NMPC ($N = 20$, 20 Hz) solves in 1.3 ms at the median and 2.2 ms at the 99th percentile ($n = 1009$), i.e. 2.6 % and 4.4 % of its 50 ms period. The deployed 16-state first-moment MHE ($N = 20$, 10 Hz) solves in 9.2 ms at the median, 24.6 ms at the 90th and 59.2 ms at the 99th percentile, with a 72.8 ms maximum ($n = 3740$, first 20 frames per flight discarded), against a 100 ms period—roughly 7 $\times$ the 1.4 ms median of the historical mass-only estimator. This meets the deadline in simulation, but the tail margin is narrow, so we make *no* onboard-headroom claim; that requires measurement on the target computer. Setpoints are published at 50 Hz by interpolating the latest prediction.

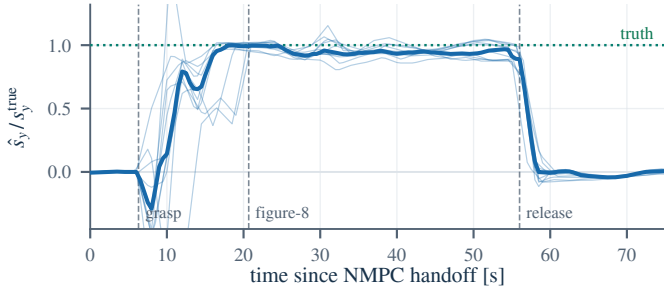


Fig. 3. First mass moment against ground truth (0.15 kg at 2 m/s, $n = 11$, each normalised by its own truth). Truth is reconstructed offline as $m_P r_y$ from the simulator-measured grasp offset and never reaches the estimator, which runs with no horizontal attachment geometry. $\hat{s}_y$ is zero before grasp, rises with no attach notification, holds near truth through the figure-eight, and decays to zero after release. Across all 31 eligible flights the carry-phase estimate is biased toward zero (slope 0.85; median magnitude underestimate 10.8%, same direction in 31/31; largest at 0.30 kg, median 15.7%). The cause of this bias is not identified; the moment-gated detector uses the ratio $\|s\|/\|s\|_{ref}$, which is insensitive to a common scale bias.

VI. RESULTS

Provenance. The lifecycle cohort uses the final 16-state first-moment interface. The baseline matrix was recorded with the earlier mass-only 14-state MHE and supports only the bounded baseline comparison stated below; it does not evaluate the proposed first-moment interface. In Table V, *truth* gives the NMPC the attachment geometry measured in simulation, *on-line* replaces the CoM offset with a motor-torque-inversion estimate, and *ll* is the L1-adaptive baseline.

A. Eventless grasp-release lifecycle

The controller issues the gripper-release command itself, but deliberately does not use that command to clear its payload model. We report every eventless flight in the final campaign that met the pre-registered eligibility criteria—successful MHE payload evidence and survival to the release command—and report the excluded attempts separately.

a) Paired command-triggered baseline: We retain the pre-registered paired design rather than selecting arm B alone after the comparison. Arm A is a command-triggered baseline on the legacy interface: it treats the release command as the detachment event and immediately resets payload geometry and inner-loop scaling. Arm B is the proposed interface. The arms also differ before release: arm A uses the legacy, non-continuous geometry path and schedules the inner loop by rewriting PX4 rate gains, whereas arm B uses the continuous $m/s/J$ frame and body-rate-command scaling. A pair enters the comparison only when both arms are eligible. Across 23 such pairs (8/7/8 across the three conditions) both arms complete 23/23 releases with no observed post-release instability (Table II); arm A has zero confirmation delay by definition.

Under delayed detachment a command-triggered baseline is expected to clear prematurely; the experiment measures this temporal mismatch and the confirmation delay that evidence-based updates incur. A fault injector received the normal release command but retained the physical DetachableJoint

TABLE II
PAIRED COMMAND-TRIGGERED ABLATION. ARM A CLEARS THE MODEL AT THE COMMAND BY CONSTRUCTION; ARM B REQUIRES ESTIMATOR EVIDENCE. THE LAST ROW INJECTS A COMMAND/PHYSICAL-DETACHMENT MISMATCH; “EARLY” MEANS MODEL CLEARING BEFORE THE DETACHABLEJOINT IS REMOVED (“-”: NO HOLD INJECTED, NOT SCORED). THE ONE VALID BUT UNPAIRED ARM B RUN OF THAT ROW ALSO DID NOT CLEAR EARLY (0/6).

condition	pairs	A compl.	B compl.	early clear	A/B
0.15 kg, 4 m/s	8	8/8	8/8	–	
0.30 kg, 4 m/s	7	7/7	7/7	–	
0.15 kg, 2 m/s	8	8/8	8/8	–	
nominal pooled	23	23/23	23/23	–	
0.15 kg, 2 m/s + 4 s hold	5	5/5	5/5	5/5 / 0/5	

for a median 4.01 s. Of twelve arm-runs, one arm A run never armed PX4, leaving five valid pairs and one valid unpaired arm B run. In all five pairs arm A cleared its payload model at the command, a median 4.01 s before physical detachment. Arm B cleared before detachment in none of its six valid runs: it entered UNRESOLVED in 5/6 and then confirmed a median 12.45 s after detachment, and the remaining run confirmed 3.00 s after detachment. All eleven valid runs had an independent Gazebo DETACHED record. Median position-error peaks during the hold were 0.071 m (A) and 0.068 m (B), so the result supports a difference in model–plant consistency, not a tracking advantage or crash avoidance. We did not test a permanently jammed gripper, under which arm A would carry a loaded vehicle with an empty model.

b) Completion and delay of the complete interface:

The complete interface confirmed release in 31/31 eligible flights with no observed post-release instability (Table III; two-sided 95 % Clopper–Pearson interval [88.8 %, 100 %]). This is a conditional closed-loop result for the interface as a whole, not a validation of any single release detector: Table IV shows that three confirmation paths occurred.

The confirmation delay is *bimodal*. In 42 % of flights the confidence crosses threshold while the vehicle is still flying the figure-eight, and confirmation completes in a median 3.46 s (Fig. 4). In the remaining 58 % no release evidence is available in maneuver— $\hat{s}$ has drifted under sustained excitation or the frozen reference never formed—so the system reaches its 12 s timeout, declares UNRESOLVED, and executes a 3 s smooth brake to hover; confirmation then completes at a median 16.48 s. Table III keeps the two modes separate.

No flight in any path declared release before the command. In 26 of the 31 flights the payload-presence machine made exactly two transitions, one at grasp and one at release; in the five confidence-only flights it never declared EMPTY within the logged window (over 40 s past confirmation). Four of the 31 load declarations were raised by the estimator’s own directional thrust-residual evidence rather than sustained mass evidence; no external grasp signal was consumed. The published first moment decayed continuously rather than stepping to zero (Fig. 3).

c) Development-stage evidence: Two earlier tests motivated the final design but used earlier code and do not validate

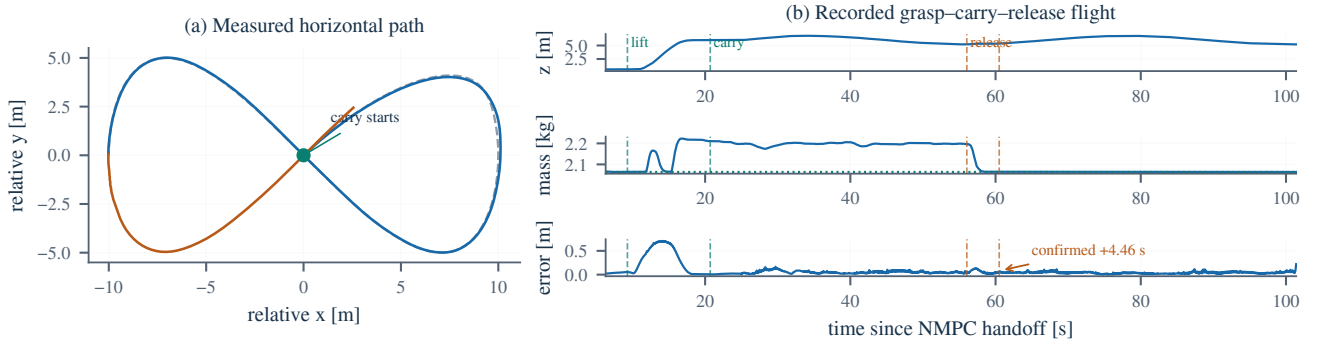


Fig. 4. One eventless flight from the final batch (0.15 kg, 2 m/s). Left: the measured horizontal path (blue while carrying, orange after release) against the nominal reference including its amplitude ramp (grey dashed). Right: recorded height, controller-consumed mass estimate, and position-error norm through grasp, lift, carrying, release command, and confirmation. This flight confirms in maneuver, 4.46 s after the command; it illustrates one execution and does not replace the cohort statistics in Table III.

TABLE III

RELEASE COMPLETION AND CONFIRMATION DELAY (ALL ELIGIBLE ARM B FLIGHTS). DELAY RUNS FROM THE RELEASE COMMAND TO THE END OF THE 1.0 s CONFIDENCE HOLD. THE TWO DELAY COLUMNS ARE DISTINCT CONTROL PATHS AND ARE NOT POOLED.

condition	n	compl.	delay median [IQR]	
			in-maneuver	via brake-to-hover
0.15 kg, 4 m/s	9	9/9	2.78 s [2.76, 2.90]	16.56 s [16.46, 16.60]
0.30 kg, 4 m/s	11	11/11	7.08 s [6.96, 7.50]	16.40 s [15.76, 16.60]
0.15 kg, 2 m/s	11	11/11	3.46 s [3.16, 4.36]	16.48 s [16.40, 16.50]
pooled	31	31/31	3.46 s [2.86, 4.46]	16.48 s [16.36, 16.60]

Overall delay max 16.70 s. Share confirming only after brake-to-hover: 56 %, 64 %, 55 % (rows in order), 58 % pooled.

TABLE IV

CONFIRMATION PATHS IN THE 31 FLIGHTS OF TABLE III. EVERY CONFIRMATION USED THE CONTROLLER'S CONFIDENCE ROUTE (§IV); COLUMNS RECORD THE ESTIMATOR DECLARATION, IF ANY. *Moment-gated*: THE DETECTOR DECLARED BEFORE THE CONFIDENCE CROSSED. *Mass-domain*: THE FROZEN REFERENCE NEVER FORMED; THE CONFIDENCE CROSSED DURING OR SHORTLY AFTER THE BRAKE TO HOVER AND THE MASS-DOMAIN ROUTE DECLARED 0.9 s to 3.1 s LATER. *Confidence only*: THE REFERENCE NEVER FORMED AND NO DECLARATION OCCURRED (ALL IN MANEUVER).

condition	n	moment-gated	mass-domain	confidence only
0.15 kg, 4 m/s	9	9	0	0
0.30 kg, 4 m/s	11	1	7	3
0.15 kg, 2 m/s	11	9	0	2
pooled	31	19	7	5

the final detector. A quantity-residual fast branch without moment evidence declared release prematurely in 3/9 randomized flights during a 4 m/s figure-eight; loaded-flight mass estimates could remain below the nominal empty threshold for up to 48 s. After its removal, a 12-flight randomized test of the moment-gated detector, run before the frozen reference was introduced, produced no pre-command declaration, 11 moment-gated releases (median detector delay 1.9 s), and one flight that stayed UNRESOLVED.

d) *Attempts and exclusions*: The paired campaign comprised 81 attempts. Arm B (proposed) made 40: 31 eligible,

TABLE V

THREE-WAY METHOD $\times$ SCENARIO MATRIX ($n = 5$): POS_ERR PEAK [M] / RECOVERY [S].

cell (mass/ecc)	truth	online	l1
0.2 / 0.05	0.806 / 4.28	0.768 / 3.70	0.821 / 4.26
0.2 / 0.10	0.794 / 3.62	0.810 / 3.92	0.812 / 4.24
0.3 / 0.05	0.834 / 3.56	0.837 / 3.72	0.883 / 4.90
0.3 / 0.10	1.011 / 3.64	0.876 / 3.88	0.891 / 4.54

six crashed before the release command, and three lacked MHE payload evidence (the logs cannot distinguish capture failure from non-recognition). Arm A made 41: 30 eligible and 11 pre-release crashes. Because the arms differ before release, these rates are reported per arm rather than pooled or compared, and neither characterizes end-to-end mission reliability. Table II uses the 23 pairs with both arms eligible; Tables III–IV use all 31 eligible arm B flights. Tail-window tracking is not compared, because arm B brakes to hover after an unresolved timeout while arm A continues the figure-eight.

B. Baselines and ablations

L1-NMPC three-way matrix (Table V). {truth, online, l1} $\times$ {0.2, 0.3} kg $\times$ {0.05, 0.10} m, $n = 5$, 60/60 runs stable, zero divergence. (i) Within-cell transient peaks agree to 0.05 m; L1 is higher in the two low-eccentricity cells ($p = .008, .012$ uncorrected, not surviving correction over eight tests), so we claim *no* transient advantage. (ii) Against the deployed *online* estimate, **L1 recovers 8 % to 32 % slower** (median 16 %; significant in three of four cells); the mechanism is that its compensation low-pass cutoff ω_c is bounded by closed-loop stability (0.5 stable, 1.0 unstable on this platform), a bandwidth wall that explicit estimation does not face. (iii) L1 yields no interpretable m/c_{xy} and still requires the same class prior for gain scaling; it is a translational 3-DoF augmentation driven by T_{phys} (the commanded thrust self-excites). The larger *truth* peak in cell 0.3/0.10 is not explained; more accurate parameters need not reduce a transient peak. All three arms used the earlier mass-only MHE.

VII. LIMITATIONS AND FUTURE WORK

All system-level evidence is presently SITL-only. Hardware migration replaces simulated motor speeds with ESC telemetry and must re-identify thrust/torque coefficients, noise floors, timing, and gripper compliance before free flight.

Thrust proxy. In simulation the plant generates rotor thrust with the same square law $k\omega^2$ that the estimator uses, and motor speeds are exact. The experiments therefore do not test thrust-map error, speed-measurement noise, bias or delay, or induced-inflow mismatch at 4 m/s [22]. Such a test must perturb the plant's thrust map independently of the estimator's; changing a shared k cannot probe it.

Missing baselines. We have not compared against a command-triggered baseline that also requires persistent residual confirmation, a fixed-parameter NMPC, a CUSUM detector [19] on the same signals under a matched false-alarm constraint, or L1-NMPC with the final estimator. The removed quantity-residual branch is not a substitute for the first: a command-triggered detector opens its window only after the command and faces different false-alarm exposure. The vertical arm r_z is unobservable from the thrust-moment channel and remains a weak prior; the s_x /pitch channel is noisier than s_y /roll. The deployed diagonal inertia approximation also drops the small $O(r_{xy}^2)$ and yaw-inertia terms.

Release-detector boundary. First-moment augmentation is now part of the default path. The development-stage tests and the three final-batch confirmation paths are reported in Sec. VI-A; 31/31 completions do not establish distribution-wide detector robustness. In 12/31 flights the healthy, low-dynamic, in-envelope reference window never filled after the LOADED declaration, so the moment-ratio test never ran. Out-of-envelope $\hat{s}$ samples are excluded from moment votes but still reach the controller's c_{xy} through filtering and slew limits. Reliable reference formation therefore remains necessary for a robust autonomous release detector.

All reported release batches explicitly enable the release-evidence path. Its built-in default is off, under which every flight ends UNRESOLVED, so deployment requires an explicit enable plus startup validation of the configuration.

First-moment drift. Sustained figure-eight excitation can increase $\|\hat{s}\|$ from 0.012 kg m to 0.031 kg m; withholding moment evidence then delays confirmation. Deceleration restored the vote within 3.3 s to 3.4 s in the affected test flights. Together with missing-reference cases, this yields the brake-to-hover delays in Table III. An online counter of first-moment envelope exceedances correlates with this outcome at 4 m/s (in-maneuver confirmation 7/8 for zero exceedances versus 1/12 otherwise; Fisher $p < 0.001$), but misses all six delayed 2 m/s cases. Earlier exit from the maneuver remains unimplemented and requires validation.

Pre-release failures. Of the six arm B crashes (Sec. VI-A), two occurred before the figure-eight at 0.15 kg/2 m/s, whereas four were loaded figure-eight divergences at 0.30 kg/4 m/s, an operating boundary of that condition. Their grasp offsets ($|r_y| = 0.07$ m to 0.14 m) and pre-maneuver roll trim reconstructed from rotor speeds (0.20 N m to 0.40 N m against

a 0.5 N m limit) lie within the range of the eleven surviving flights at that condition, so attachment-induced torque saturation does not explain them; their mechanism remains unidentified. We make no claim about grasp, lift, figure-eight stability, or end-to-end mission reliability.

Self-triggered de-weighting. The residual-triggered β_j schedule of (3) fired in 19/21 eligible 0.30 kg flights and 2/18 at 0.15 kg/4 m/s (both arms), but in none of the 22 at 2 m/s. Its 4.4 N threshold exceeds both payload weights, so it responds to fast thrust transients rather than discriminating payload weight; in these flights it was crossed by grasp and release transients. A later repeat of the two 4 m/s conditions with it disabled ($\beta_j \equiv 1$; not interleaved with the original runs) completed 20 of 21 eligible eventless releases. The moment-gated detector again declared first in 10 of them, and no load declaration relied on the residual trigger (4 in the original runs). The incomplete flight is a failure we report rather than exclude: during the loaded figure-eight the MHE re-initialized repeatedly after consecutive solver failures, and the vehicle diverged 2.8 s after the release command, before the UNRESOLVED timeout. Re-initialization seeding is under separate investigation.

A sustained vertical up/downdraft remains intrinsically confounded with a payload change on a collective-thrust-only residual; wind is not evaluated in the reported cohort.

Simulator fidelity. (i) gz-sim Harmonic cannot change a link's inertia at runtime in a way the physics adopts (issue #2733), so every payload change is produced by the DetachableJoint platform, which attaches a separate rigid body (§V). (ii) The payload is rigidly attached; gripper compliance and pendulum modes are not modeled. Neither affects which signals the estimator consumes.

VIII. CONCLUSION

We presented an eventless payload-model interface for quadrotor NMPC that estimates composite mass and the horizontal first mass moment from odometry and a rotor-speed thrust proxy, and that clears its payload model only on persistent estimator evidence, never on its own release command. In PX4 SITL the complete interface confirmed release in all 31 eligible flights (31 of 40 attempts) through three confirmation paths, with median delays of 3.46 s in maneuver and 16.48 s after braking to hover. Under a 4 s delayed detachment, a command-triggered baseline cleared its model early in 5/5 pairs and the proposed interface in 0/5. Hardware validation, thrust-map mismatch, reliable moment-reference formation, and stronger baselines remain open.

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